Handoff PDF generated: 2026-05-27 22:47

Handoff —¿—°—”—µÑ: Recognition of cash receipts

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— Ñ—¾—µ—”Ñ: Telegramâ ———¾Ñ—·—»Ñ Ñ—°Ñ—¿—¾—•—‰—°—†—°—‰—‚Ñ —”—°ÑÑ—¾—†ÑÑ Ñ—µ—”—¾—† —‚ —•—°—

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Project Overview

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— Ñ—‰—¾—†—‰Ñ—µ—”—¾…—¿—¾—‰—µ—‰ÑÑ
- bot.py: Telegram UI/—»—¾—‡—‚—”—°, —¿Ñ—‚Ñ—… Ñ—¾Ñ—¾Ñ—µ¶—‚—…Ñ â —¾·—‰—¾Ñ—¾Ñ—¾â / â —·—»—‚—‰
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Ñ—¾…—‰—µ—‰—‚ÑÑ, —…—µÑ—°—·—°—‰—‰Ñ—µ_meta.
- receipt_qr.py: ÑÑ—µ—‰—‚—µQR (QReader —¿Ñ—‚, —‰—°—»—‚Ñ—‚—,, —‚—‰—°Ñ—µOpenCV) + —¿—°ÑÑ—‚—‰—‡ Ñ
—⁄— —¡.
- system_prompt.txt: Ñ—‚ÑÑ—µ…—‰Ñ—„ —¿Ñ—¾…—¿Ñ (ÑÑÑ—¾‡—‚—„ JSON, —¿—¾—»—‰—¾Ñ—° —¿—¾—•—‚Ñ—‚,
Ñ—°—…—¾—¿Ñ—¾—†—µÑ—”—°).
- google_sheets.py: —•—°—¿—‚ÑÑ —† Google Sheets + —†—°—»—‚—·—°Ñ—‚—‚/—¿Ñ—µ·Ñ—¿Ñ—µ¶—·—µ—‰—‚Ñ, —”

— —¾Ñ—¾—” —¾—Ñ—°———¾Ñ—”—‚, —¾·—‰—¾—‡—¾Ñ—µ—”—°
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(Ñ—¾—»Ñ—”—¾·—»Ñ —¾·—‚—‰—¾Ñ—‰—¾‡—¾—”—°—·Ñ—°).
2. —⁄—‚Ñ—”—°—»Ñ—‰Ñ—„ QR (—¿Ñ—‚—¾Ñ—‚Ñ—µÑ):
   - — —µ—”—¾·—‚Ñ—¾—†—°—‰—‚—µQR: QReader (—µÑ—»—‚, ÑÑÑ—°—‰—¾—†—»—µ—‰ —, Ñ—°———¾Ñ—°—µÑ) â —
   - — Ñ—»—‚, ÑÑÑ—¾—”—° Ñ—¾—¾Ñ—†—µÑÑÑ—†Ñ—µÑ Ñ—¾Ñ—…—°ÑÑ —⁄— —¡ (t, s, fn, i|fd, fp), —‚—•—†—»—µ—”—°ÑÑÑÑ: —
—⁄— (—”—°—” â —‰—¾…—µÑ Ñ—µ—”—°â), fp, —‚Ñ—¾‡—¾—†—°Ñ ÑÑ—…—…—°.
3. OpenAI Vision (ocr.py):
   - —¡—‚ÑÑ—µ…—‰Ñ—„ —¿Ñ—¾…—¿Ñ —‚—• system_prompt.txt.
   - Userâ —‚—‰ÑÑÑÑ—”Ñ—‚Ñ —‚—• build_user_text() (—·—»Ñ —·—»—‚—‰—‰ÑÑ Ñ—µ—”—¾† â —¿Ñ—¾—¿—¾—»—‰
   - — Ñ—»—‚, QR Ñ—°Ñ—¿—¾—•—‰—°—‰ â —† userâ Ñ—µ—”ÑÑ —·—¾———°—†—»Ñ—µÑÑÑ —”—¾—‰Ñ—µ—”ÑÑ Ñ ÑÑ—°
4. —¡—»—‚Ñ—‰—‚—µÑ—µ—•Ñ—»ÑÑ—°Ñ—°:
   - — —¾—»Ñ —‚—• QR —¿Ñ—‚—‰Ñ—·—‚Ñ—µ—»Ñ—‰—¾—¿—°ÑÑ—°ÑÑÑ —† receipt_info
(date/time/receipt_number/fiscal_sign/total_amount/fiscal_storage_number).
5. — —°—»—‚—·—°Ñ—‚Ñ / —¿—¾†Ñ—¾Ñ:
   - should_recheck() Ñ—µÑ—°—µÑ, —‰Ñ—¶—‰—° —»—‚, 2â Ñ—¿—¾¿ÑÑ—”—° LLM.
   - _meta.needs_review —†ÑÑÑ—°—†—»Ñ—µÑÑÑ —¿—¾Ñ—‚—‰—°—»Ñ—‰—¾…Ñ JSON —¿—¾Ñ—»—µ—¿—¾¿ÑÑ—¾
6. — —°—¿—‚ÑÑ —† Sheets:
   - — —‚ÑÑ Ñ—µ—”—¾†: Aâ N ÑÑ—°—‰—·—°ÑÑ—‰Ñ—µ—¿—¾—»Ñ, O = —⁄— (—‚—• QR), P = QR Ñ—°Ñ—¿—¾—•—‰—
   - — —‚ÑÑ Ñ—¾†—°Ñ—¾†: —¿—¾—•—‚Ñ—‚—‚, items.

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Quickstart (ÑÑ—¾—Ñ —·ÑÑ—‡—¾—„ —‚—‰—¶—µ—‰—µÑ —•—°—¿ÑÑÑ—‚—» —•—° 10 —…—‚—‰ÑÑ)

— Ñ—µ—·ÑÑ—»—¾—†—‚Ñ
- Python 3.10+
- — —¾ÑÑÑ—¿ —”:
  - Ñ—¾—”—µ—‰Ñ Telegramâ ———¾Ñ—° (BotFather)
  - —”—»ÑÑÑ OpenAI
  - Google Spreadsheet —‚ JSONâ —”—»ÑÑÑ Service Account (Ñ —·—¾ÑÑÑ—¿—¾… —” Ñ—°———»—‚Ñ—µ)

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— —¾—‰Ñ—‚—‡ÑÑ—ºÑ—‚Ñ
1. —¡—”—¾—¿—‚Ñ—¾—†—ºÑÑ .env.example â .env
2. — —°—¿—¾—»—‰—‚ÑÑ —…—‚—‰—‚—…Ñ—…:
   - TELEGRAM_BOT_TOKEN
   - OPENAI_API_KEY
   - OPENAI_VISION_MODEL (—‚—»—‚ OPENAI_MODEL)
   - GOOGLE_SHEETS_SPREADSHEET_ID
   - GOOGLE_SERVICE_ACCOUNT_FILE (—‰—°—¿Ñ—‚—…—µÑ credentials.json, Ñ—°—„—» —‰—µ—”—¾…—…—‚Ñ—

— —¿Ñ—‚—¾—‰—°—»Ñ—‰—¾
- RECEIPT_QR_ENABLED=0 â —¾Ñ—”—»ÑÑ—‚ÑÑ ÑÑ—°—¿ QR (fallback: Ñ—¾—»Ñ—”—¾ vision OCR)
- RECEIPT_PREPROCESS_* â ÑÑ—‰—‚—‰—‡ —¿Ñ—µ—·—¾—Ñ—°———¾Ñ—”—‚ —‚—•—¾—Ñ—°—¶—µ—‰—‚—„
- RECEIPT_STATE_BACKEND=sqlite|redis â ÑÑ—°—‰—‚—»—‚Ñ—µÑ—¾ÑÑ—¾Ñ—‰—‚Ñ —•—°—‡ÑÑ—•—”—‚ Ñ—µ—”—°

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Prompt Engineering Guide (Ñ—µÑ—‰—¾—†—‚—”)

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1) system_prompt.txt (—¾Ñ—‰—¾—†—‰—¾—„ Ñ—‚ÑÑ—µ—…—‰Ñ—„ —¿Ñ—¾…—¿Ñ)

— —¾—»Ñ —‚ —•—°—·—ºÑ—°
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- —¢Ñ—µ——Ñ—µÑ ÑÑÑ—¾—‡—‚—„ JSON —‚ —¿—¾—»—‰—¾ÑÑ (—‰—µÑ—µÑÑÑÑ ÑÑÑ—¾—”—‚„ —»ÑÑÑ—µ—•—°—·Ñ———

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- — Ñ—»—‚ QRâ —”—¾—‰Ñ—µ—”ÑÑ —¿—µÑ—µ—·—°—‰: ÑÑ—‚Ñ—ºÑÑ —¿—¾—»Ñ —‚—• QR ÑÑ—°—»—¾—‰—‰Ñ—…—‚,
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ÑÑ—µ———¾—†—°—‰—,Ñ —¿—¾—»—‰—¾ÑÑ.

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2) Userâ —,—‰ÑÑÑÑ—”Ñ—,Ñ—† ocr.py::build_user_text() (—¾—¿—µÑ—°Ñ—,—†—‰—°Ñ —¿—¾·Ñ—”—°—•—”—°)

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  - —¿—¾†Ñ—¾Ñ—‰—°Ñ —¿—¾—¿ÑÑ—”—° (attempt â¥ 2) ÑÑ—,—»—,—†—°—µÑ Ñ—°—…—¾—¿Ñ—¾†—µÑ—”Ñ—, —°Ñ—
  - —¿Ñ—, —‰—°—»—,Ñ—,—, QR —·—¾———°—†—»Ñ—µÑÑÑ qr_context_for_prompt().

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- —¢—°———»—,Ñ—° (—»—,ÑÑ Ñ—µ—”—¾†):
  - O (—⁄— —,—• QR): —•—°—¿—¾—»—‰—µ—‰—¾â QRâ payload —⁄— —¡ Ñ—°Ñ—¿—¾—•—‰—°—‰ —, —¿Ñ—,—…—
  - P (QR Ñ—°Ñ—¿—¾—•—‰—°—‰): â — —°⁄— —µÑâ —¿—¾_meta.qr.used.
- — —”—¾·—µ/—»—¾‡—,—”—µ
  - receipt_qr.py::extract_fiscal_data_from_receipt_images() —‰—°ÑÑ—» QR —,
parse_fiscal_qr_payload() —†—µÑ—‰Ñ—» patch.
  - ocr.py —†Ñ—¿—¾»—‰—,—» apply_qr_patch_to_receipt() (QR —†Ñ—µ—‡—·—° —,—…—µ—µÑ —¿Ñ—,—¾Ñ—,Ñ—µÑ—·

— —·—µÑ—…—¾ÑÑ—µÑÑ —¿Ñ—,Ñ—,—‰Ñ —¿Ñ—µ·Ñ—¿Ñ—µ¶—·—µ—‰—,—„
- — Sheets:
  - validation_status / validation_comment Ñ—¾Ñ—…—,ÑÑÑÑÑÑ—† google_sheets.py.
- — runtime:
  - _meta.needs_review —†ÑÑÑ—°—†—»Ñ—µÑÑÑ—† ocr.py —¿—¾Ñ—,—‰—°—»Ñ—‰—¾…Ñ Ñ—µ•Ñ—»ÑÑ—°ÑÑ (—¿—
QRâ —¿—°ÑÑ—°).

— Ñ—»—, â —¿—¾•—,Ñ—,—, ÑÑ—µÑ—°—»—,â (Ñ—µ—‰Ñ/—”—¾»-—†—¾¿—µÑ—µ¿ÑÑ—°—»—,ÑÑ)
— Ñ—¾†—µÑÑÑÑ —¿—¾¿—¾ÑÑ—·—”Ñ:
1. —”—°Ñ—µÑÑ—†—¾,ÑÑ—¾·—‰ÑÑ —”—°—·Ñ—¾† (—¿Ñ—°—†Ñ—„ —”Ñ—°—„, Ñ—‡—»Ñ, ———»—,—”—,);
2. —¿Ñ—µ·—¾Ñ—°———¾—”Ñ (RECEIPT_PREPROCESS_*, —¾Ñ—¾———µ—‰—‰—¾crop);
3. —¿Ñ—°—†—,—»—° â —”—°—” ÑÑÑ—¾,ÑÑ —¿—¾•—,Ñ—,Ñâ —† system_prompt.txt;

4. ocr.py::should_recheck() â —‰—µÑ—»—,Ñ—”—¾—… —»—, —…Ñ—‡—”—¾—¶ÑÑÑ—”—¾Ñ—µÑ—°—µÑ —¿Ñ—¾retry.

— Ñ—»—, â Ñ—µ—” —·Ñ———»—,ÑÑ—µÑÑÑâ —,—»—, â —‰—µ—•—°—¿—,ÑÑ—†—°—µÑÑÑâ
- — —°Ñ—,Ñ—° —¾Ñ— —·Ñ———»—µ—„ —† google_sheets.py —,Ñ—¿—¾—»Ñ—•Ñ—µÑ composite key:
  - receipt_number | date | total_amount
  - — —°—¶—‰—¾—¿Ñ—, ÑÑ—¿—µÑ—‰—¾—… QR, receipt_number ÑÑ—¾—⁄— (i/fd), —° total_amount â —,—• QR.

— Ñ—»—, —…—µ—‰Ñ—µ—… ÑÑ—µ—…Ñ Ñ—°———»—,ÑÑ
- — —,ÑÑ Ñ—µ—”—¾—†: Ñ—µ—„Ñ—°Ñ —•—°—¿—,ÑÑ—†—°—µ—… —·—,—°—¿—°—•—¾—‰ A:P.
  - — —¾———°—†—»—µ—‰—,—µ—‰—¾—†ÑÑ —”—¾—»—¾—‰—¾—” ÑÑ—µ——Ñ—µÑ —¿Ñ—°—†—”—, google_sheets.py —
README.md/HANDOFF.md.
- — —,ÑÑ Ñ—¾—†—°Ñ—¾—†: Ñ—¾Ñ—…—,ÑÑ—µÑÑÑ —¾Ñ—·—µ—»Ñ—‰—¾—¿Ñ—¾—†—µÑÑÑÑ Ñ—¾Ñ—…Ñ—»Ñ —,Ñ—¾—‡—°

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Troubleshooting FAQ (——ÑÑÑÑÑ—µ—¾Ñ—†—µÑÑ)

1) — —¾Ñ —‰—µÑÑ—°ÑÑÑ—µÑ: ModuleNotFoundError: No module named ’telegram’
— Ñ—,Ñ—,—‰—°: —•—°—¿ÑÑ—” —‰—µ—,—• —†—,ÑÑÑ—°—»Ñ—‰—¾—‡—¾—¾”ÑÑ—¶—µ—‰—,Ñ, —‡—·—µÑÑ—¾ÑÑ —
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- —°—”Ñ—,—†—,Ñ—¾—†—°ÑÑ .venv —, —•—°—¿ÑÑ—”—°ÑÑ python bot.py, —,—»—,
- —•—°—¿ÑÑ—”—°ÑÑ —‰—°—¿ÑÑ—…ÑÑ .\.venv\Scripts\python.exe .\bot.py.

2) — —°—·—°—µÑ Ñ TELEGRAM_BOT_TOKEN —‰—µ—•—°—·—°—‰ / OPENAI_API_KEY —‰—µ—•—°—·—°—‰
— Ñ—,Ñ—,—‰—°: —‰—µ—•—°—¿—¾—»—‰—µ—‰ .env —,—»—, —¿—µÑ—µ—…—µ—‰—‰Ñ—µ—¾”ÑÑ—¶—µ—‰—,Ñ
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- —•—°—¿—¾—»—‰—,ÑÑ TELEGRAM_BOT_TOKEN, OPENAI_API_KEY;
- —¿Ñ—, —‰—µ—¾—Ñ—¾—·—,—…—¾ÑÑ—, —¿—µÑ—µ—•—°—¿ÑÑÑ—,ÑÑ —¿Ñ—¾Ñ—µÑÑ ———¾Ñ—°.

3) Google Sheets —‰—µ—¿—,Ñ—µÑ: FileNotFoundError credentials.json / —¾Ñ—,———”—, —°—†Ñ—¾Ñ—,—•—°Ñ—,—,
— Ñ—,Ñ—,—‰—°: —‰—µ—†—µÑ—‰Ñ—„ —¿ÑÑÑ GOOGLE_SERVICE_ACCOUNT_FILE, —¾ÑÑÑÑÑÑ—†Ñ—µÑ Ñ—°—„—» —
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- Ñ———µ—·—,ÑÑÑÑ, ÑÑ—¾GOOGLE_SERVICE_ACCOUNT_FILE Ñ—”—°—•Ñ—†—°—µÑ —‰—° ÑÑÑ—µÑÑ—†ÑÑÑ—,—„ JSO
- —†Ñ—·—°ÑÑ Ñ—°———»—,Ñ—µ—·—¾ÑÑÑ—¿ —‰—° eâ mail Ñ—µÑ—†—,Ñ—‰—¾—‡—¾—°—”—”—°Ñ—‰Ñ—° (—,—• JSO
- —¿Ñ—¾—†—µÑ—,ÑÑ GOOGLE_SHEETS_SPREADSHEET_ID —, —,—…—µ—‰—° —»—,ÑÑ—¾—†.

4) QR â —‰—µÑ—°Ñ—¿—¾—•—‰—°—‰â (P=— —µÑ), Ñ—¾ÑÑ —‰—° Ñ—µ—”—µQR —µÑÑÑ
— Ñ—,Ñ—,—‰Ñ:
- QR —‰—° Ñ—¾Ñ—¾Ñ—»—,Ñ—”—¾—… —…—µ—»—”—,—„/Ñ—°—•—…ÑÑÑ—„/—† ———»—,—”—°Ñ;
- QR —¿Ñ—¾Ñ—,Ñ—°—»ÑÑ, —‰—¾ÑÑÑ—¾”—° —‰—µ—¿—¾Ñ—¾¶—° —‰—° Ñ—,Ñ—”—°—»Ñ—‰Ñ—„ Ñ—¾Ñ—…—°Ñ —
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- Ñ—»ÑÑÑ—,ÑÑ Ñ—¾Ñ—¾———»—,—¶—µ ———µ—• ———»—,—”—¾—†, Ñ—µ—•—”—¾ÑÑÑ;
- —†Ñ—µ—…—µ—‰—‰—¾Ñ—”—»ÑÑ—,ÑÑ —°—‡Ñ—µÑÑ—,—†—‰ÑÑ —¾Ñ—°———¾—”Ñ (—µÑ—»—, —†—”—»ÑÑ—°
- —¿Ñ—¾—†—µÑ—,ÑÑ, ÑÑ—¾RECEIPT_QR_ENABLED=1;
- —¿Ñ—, —‰—µ—¾—Ñ—¾—·—,—…—¾ÑÑ—, —·—¾———°—†—,ÑÑ —¿—¾—·—·—µÑ—¶—”Ñ —µÑÑ —¾—·—‰—¾—‡—¾—†—
receipt_qr.py::parse_fiscal_qr_payload().

5) QR Ñ—°Ñ—¿—¾—•—‰—°—‰, —‰—¾—¿—¾—»Ñ—† Ñ—°———»—,Ñ—µâ ÑÑÑ—°—‰—‰Ñ—µâ (—·—°Ñ—°/ÑÑ—…—…
— Ñ—,Ñ—,—‰—°: —† ÑÑ—¾—… —¿Ñ—¾—µ”Ñ—µQR —·—»Ñ —”—»ÑÑ—µ—†ÑÑ —¿—¾—»—µ—„ â —,ÑÑ—,—‰—° —, —,—
— —µÑ—µ—‰—,—µ
- Ñ—†—µÑ—,ÑÑ â ÑÑÑ—¾—„ payloadâ (—† _meta.qr.raw, —µÑ—»—, —†Ñ—†—¾—·—,ÑÑÑ/—»—¾—‡—,ÑÑ—µÑÑÑ) —, —•—‰—
- —µÑ—»—, —‰—° —”—¾‰—”Ñ—µÑ—‰—¾—… Ñ—,—¿—µÑ—µ—”—¾—† QR —”—¾—·—,ÑÑ—µÑ ÑÑ—…—…Ñ —† —·ÑÑ—
_amount_to_total_amount_str() —† receipt_qr.py.

6) â — Ñ—µ—·Ñ—¿Ñ—µ—¶—·—µ—‰—,—µ—¿—¾ÑÑ—…—…—°—…â , Ñ—¾ÑÑ —†—,—•Ñ—°—»Ñ—‰—¾—†ÑÑ —‰—¾Ñ—…..
— Ñ—,Ñ—,—‰Ñ:
- items_total - discounts —‰—µÑÑ—¾—·—,ÑÑÑ Ñ total_amount (—¾Ñ—,———”—° ÑÑ—µ—‰—,Ñ Ñ—,Ñ—µ—», —¿Ñ—¾—¿ÑÑ—”
—·Ñ———»Ñ);
- —†—”—»ÑÑ—,—»ÑÑ needs_review —¿—¾Ñ—»—µÑ—,—‰—°—»Ñ—‰—¾—„ —¿Ñ—¾—†—µÑ—”—,.